

HUJI Applicative Research Report — Agriculture & Food — Week 34, 2026

1 paper(s) · Agriculture & Food · Weekly · Week 34, 2026 · Generated August 17, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Our latest AgriTech and FoodTech research highlights innovative approaches to food preservation and quality enhancement. One significant breakthrough focuses on leveraging nutritional interventions in livestock to directly improve end-product quality. This promising development offers a clean-label method to extend the shelf-life and visual appeal of premium beef products, addressing key challenges in the meat industry.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

• Oren Tirosh — AgriTech, FoodTech

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Vitamin E supplementation moderates salt-induced oxidation status in beef. 23/50

"New Vitamin E feed additive prevents salt-induced discoloration, extending shelf-life for premium beef products."

Main researcher: Oren Tirosh · Oren.tirosh@mail.huji.ac.il.

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ABOUT THIS RESEARCH

Researchers found that supplementing calves with high doses of Vitamin E before slaughter protects beef from oxidation and discoloration caused by salt treatments. The study establishes a specific dosage and timeframe to maintain meat quality and color during processing.

COMMERCIAL ANGLE

Development of specialized livestock feed additives or precision nutrition protocols to extend the shelf-life and visual appeal of premium salted beef products.

COMMERCIALISATION METRICS

Novelty

3/10

The research applies a well-known antioxidant (Vitamin E) to a known problem (lipid peroxidation) using standard dosage optimization; it lacks a novel mechanism, a proprietary delivery system, or a groundbreaking discovery.

Commercial Potential

6/10

The study identifies a clear, practical application for enhancing meat shelf-life and quality via a specific dosing regimen, but it lacks high-margin protectability as it utilizes a generic, well-known supplement (Vitamin E).

Market Size

6/10

The target market is the global beef industry, which is vast; however, this is an incremental improvement using a known additive (Vitamin E) rather than a disruptive platform technology.

Tech Readiness

6/10

The research has moved beyond basic proof-of-concept to in vivo optimization of dosing and duration in calves, demonstrating clear industrial applicability. However, it remains at a pilot/experimental stage and requires scale-up validation in commercial feedlots before being market-ready.

IP Strength

2/10

The research applies well-known nutritional concepts (Vitamin E supplementation) to a common problem, lacking a novel molecule, proprietary delivery system, or unique mechanism that would meet the non-obviousness threshold for strong patentability.

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